

Feature 1 (generate study guide based on weak topics):

Problem: After students take a quiz, the app tells them where they went wrong, but they have to go back to the complete study guide, hunt down the things they were wrong about, and review them, or they might not even be able to correctly identify the areas where they made mistakes.

Target students: STEM students who want to improve on the mistakes they made on the quiz.

Proposed solution: If we provide a way for them to get a study guide based on the areas they got wrong on the quiz, students are more likely to review them and improve, since they do not have to search for those topics in the main study guide on their own.

Goals:

- **Primary metric:**
 - The percentage scored on the total quiz is greater after they reviewed the study guide based on the weak areas.
- **Guardrails:**
 - The score doesn't drop below what it was when they first took the quiz.

Scope of the solution: Let students generate a personalized study guide based on the weak areas from the quiz they took.

Out of scope: A generic study guide that is not based on the things they did wrong.

MVP / Functional Requirements

Core flow:

- [P0] After quiz submission, the system tags each incorrectly answered question with its topic and marks topics with wrong answers as "weak topics."
- [P0] If weak topics exist, a "Generate study guide from weak areas" option appears below the quiz results.
- [P0] On click, the system retrieves indexed material chunks limited to the tagged weak topics.
- [P0] The system generates and displays a study guide scoped to those weak topics only.

Edge case handling:

- [P0] If the student scores perfectly, the generate-guide option does not appear; a positive confirmation state is shown instead.
- [P1] If exactly one weak topic exists, the system generates a condensed single-topic summary instead of a full guide.
- [P1] If the entire quiz is answered incorrectly, the student is redirected to the original full study guide instead of a generated one.
- [P1] If generation fails, the system retries once automatically; on repeated failure, it falls back to serving relevant material directly from the original guide.

Feedback/telemetry:

- [P0] Each generated guide includes a lightweight "was this guide helpful / on-topic?" flag, visible to the admin.
- [P0] Admin can monitor flag rate per generation batch; tagging logic is flagged for review if flag rate exceeds 10%.